

round understanding

Unit 4 · Basic Geometry · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5A
Unit 4 + Modern Education 5A Unit 14

Core traps: □ T4 Geometry - Confusing radius and diameter · $d=2r$ Reverse relationship · Incorrect
compass opening setting

SSPA association: ● High Frequency Sub-test paper 2 Geometry Basics, there must be circle identification
questions every year

Prerequisite knowledge: ● Master the relationship $d = 2r$ ● Be proficient in drawing circles with a
compass ④ Understand the relationship between circles and other shapes

Our goals: ● Understand the center, radius, and diameter of a circle ● Master the relationship $d = 2r$ ● Be
proficient in drawing a circle with a compass ④ Understand the relationship between a circle and other
figures

Student Name:

Class:

Date:

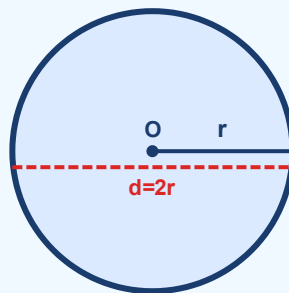
Time Spent:

I. Warm-Up Questions (total 5 question · 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	A square has a side length of 6 cm, what is its perimeter?	Basic	
2	A rectangle is 8 cm long and 5 cm wide. What is its perimeter?	Basic	
3	Which of the following graphs is not composed of curves? A. Circle B. Oval C. Square D. All of the above faces	Basic	
4	Measure the length of the following line segment AB (unit: cm) 	Basic	
5	In life, give three examples of circles and briefly explain why they are round . Examples of circles and a brief explanation of why they are round.	Advanced	

II. Core Knowledge + Worked Examples

🕒 The structure of a circle - center (O), radius (r), diameter (d)



The center O is the center point of the circle. Radius r is the distance from the center of the circle to the circumference. The diameter $d=2r$ passes through the center of the circle and connects both ends of the circle. Red dashed line = diameter.

circle key relationship

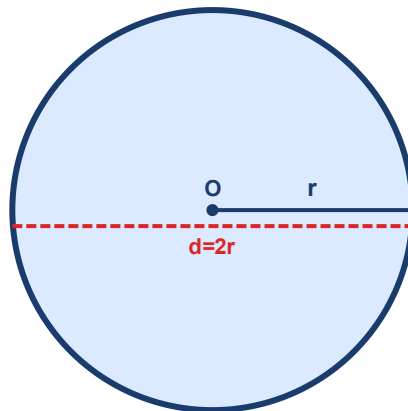
- ① **Diameter** = $2 \times \text{Radius}$ ($d = 2r$)
- ② **Radius** = $\text{Diameter} \div 2$ ($r = d \div 2$)
- ③ **Circumference** = $\pi \times \text{Diameter}$ = $2\pi r$ ($C = \pi d = 2\pi r$)
- ④ **Circle Area** = $\pi \times \text{Radius}^2$ ($A = \pi r^2$)

Knowledge point 1: Center · Radius · Diameter ($d = 2r$) 🕒 SSPA

- ① **Center of circle (O)** |||SEP|||: The exact center of the circle. The distance from every point on the circle to the center of the circle is equal. Radius (r) |||SEP|||: The distance from the center of the circle to any point

on the circumference. All radii within the same circle are equal.

- ② **Diameter (d)** |||SEP|||: A line segment that passes through the center of the circle and connects two points on the circumference. |||SEP|||, that is, the diameter is twice the radius. Chord |||SEP|||: A line segment connecting any two points on the circumference. The diameter is the longest chord
- ③ **in the circle (because it passes through the center of the circle).** ⑤ In the same circle, all diameters have the same length; $\text{diameter} = \text{radius} \times 2$; $\text{radius} = \text{diameter} \div 2$. $d = 2r$, that is, the diameter is twice the radius.
- ④ **string**: A line segment connecting any two points on the circumference. The diameter is inside the circle **longest string** (Because it passes through the center of the circle).
- ⑤ In the same circle, all diameters have the same length; $\text{diameter} = \text{radius} \times 2$; $\text{radius} = \text{diameter} \div 2$.



❑ Trap detonation example - relationship between radius and diameter

Example 1: The radius of a circle is 5 cm, what is its diameter?

✗ **Common mistakes (40% students)**

$$d = 5 \div 2 = 2.5 \text{ cm}$$

Put $d = r \div 2$, reversing the relationship

✓ **Correct solution**

$$d = 5 \times 2 = 10 \text{ cm}$$

The diameter is twice the radius: $d = 2r$

example

Example 2: The diameter of a circle is 18 cm, what is the radius?

🧠 **Tip: "Pull the diameter through the center, and reduce the radius by half. Remember $d = 2r$ clearly, $r = d \div 2$, there is no mistake!"**

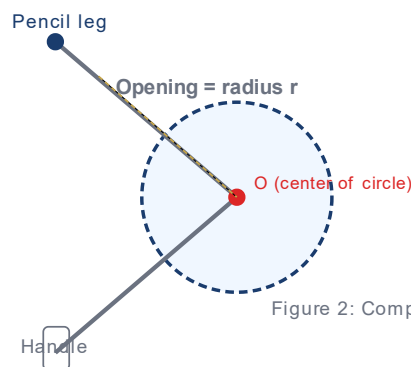
⚠ The most frequent error: reversing the relationship between radius and diameter. Remember: diameter is longer than radius! $\text{Diameter} = \text{Radius} \times 2$!

⚠ The second most frequent error: the units are not uniform. If $r = 3.5 \text{ cm}$, $d = 7 \text{ cm}$ (not 7 mm or 70 mm).

#	Question	Difficulty	Working Space
6	Radius $r = 4$ cm, diameter $d = ?$		
7	Radius $r = 7.5$ cm, diameter $d = ?$		
8	Diameter $d = 15$ cm, radius $r = ?$		
9	Diameter $d = 2.4$ cm, radius $r = ?$ (Answer in decimals)		
10	A circle has two radii OA and OB. OA = 6 cm, what is the length of OB? Why?		

Knowledge point 2: Drawing a circle - the use of compasses SSPA implementation

- ① **The two legs of the compass** are: one leg is the **pin** (fixed at the center of the circle), and the other leg is the **pencil leg** (draw the circumference).
- ② **Compass opening distance = radius** (not diameter! This is the most frequent error).
- ③ If the diameter is given in the question, you must first calculate **$r = d \div 2$** and then adjust the compass opening.
- ④ Steps to draw a circle: ① Determine the center of the circle ② Measure the radius and adjust the opening of the compass ③ Pin the center of the circle with the pin ④ Draw the circumference by going around with the foot of the pencil.
- ⑤ When drawing a circle, keep the compass vertical to the paper and do it in one go. Do not change the opening midway.



example

Example 3: Use a compass to draw a circle with a radius of 3 cm. How high should the compass opening be adjusted?

example

Example 4: Use a compass to draw a circle with a diameter of 8 cm. What should the compass opening be? Write out the complete steps.

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
11	Draw a circle with radius 2.5 cm, compass opening = ?		
12	Draw a circle of diameter 6 cm with compass opening = ? r must be calculated first.		
13	To draw a circle with a diameter of 14 cm, to what extent should the opening of the compass be adjusted?		
14	Draw a circle after adjusting the opening of the compass to 5 cm. Of this circle: (a) Radius = ? (b) Diameter = ?		

Knowledge point 3: The relationship between circles and other shapes

SSPA Advanced

① **Square inscribed in a circle** |||SEP|||: The four vertices of the square are all on the circumference.

Diagonal of the square Diagonal = diameter of the circle
Circle inscribed in the square |||SEP|||: The circle touches all four sides of the square. °

② **diameter of a circle = side length of a square** The circumferential angle on the diameter is a right angle
 |||SEP|||: A triangle with the diameter as the hypotenuse and the vertex on the circumference must be a right triangle. **The vertical line segment from the center of the circle to the chord must bisect the chord** °

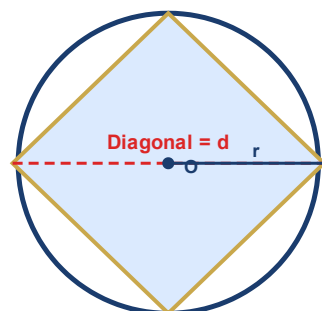
③ **(symmetry property)**. ⑤ The largest circle that can be drawn within a rectangle: diameter = shorter side (width), radius = width $\div$ 2.

④ **The vertical line segment from the center of the circle to the chord must bisect the chord** (Symmetry property).

⑤ The largest circle that can be drawn within a rectangle: diameter = shorter side (width), radius = width $\div$ 2.

Circle inscribed in a square

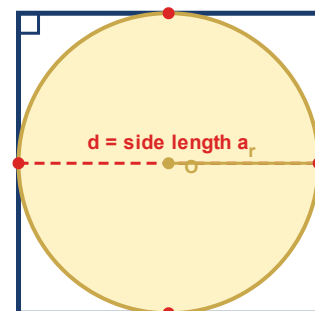
The vertex of the square is on the circumference



Diagonal = Diameter = $2r$

Square inscribed in the circle

The circle touches the four sides of the square



side length a = diameter $d = 2r$

example

Example 5: The diameter of a circle is 10 cm. What is the diagonal length of the inscribed square of this circle?

example

Example 6: The side length of a square is 8 cm. Draw the largest circle inside. What is the radius of this circle?

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
15	The diameter of a circle is 12 cm. What is the diagonal length of the square inscribed in it?		
16	The side length of a square is 6 cm. Draw the largest circle within the square. What is the diameter of the circle?		
17	A rectangle is 10 cm long and 6 cm wide. Draw the largest circle within the rectangle. The circle's: (a) diameter = ? (b) Radius = ?		
18	A triangle has as its base the diameter AB of a circle and its third vertex C on the circumference. What is the angle ACB? Why?		

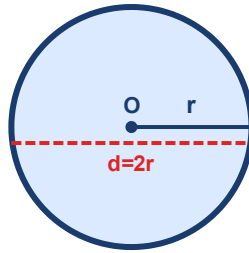
Knowledge point 4: Circle in life Application

① **Why are wheels round?** The distance from the center of the circle to the ground is constant (=radius), and the height of the axle remains unchanged when driving, making it smooth and not bumpy. Square wheels make the car bounce up and down.

② **Clock hands** |||SEP|||: **The pointer starts from the center of the circle, and the trajectory traveled by the needle tip is circular. Pointer length = radius of circle.** CD/Coin/Coaster |||SEP|||: Round shape without sharp corners, easy to make, carry and use.

③ **Why are manhole covers round?** The circular manhole cover will not fall into the wellhead in any direction (the diameter is the same), and the circle is evenly stressed.

④ **Why are manhole covers round?** The circular manhole cover will not fall into the wellhead in any direction (the diameter is the same), and the circle is evenly stressed.



example

Example 7: The minute hand of the clock is 12 cm long. What shape is drawn by moving the tip of the minute hand in one circle? What is the radius of this shape?

example

Example 8: A circular pool, the distance from the edge of the pool to the center is 5 m. A worker wants to pull a rope from one side of the pool to the other side directly opposite (through the center). What is the minimum length of the rope?

Knowledge Point Four Synchronization Practice

#	Question	Difficulty	Working Space
19	In a circular garden bed, a gardener has to walk 3 m from the center to the edge of the garden bed. What is the diameter of the flower bed?		
20	The second hand of the clock is 8 cm long. What is the radius of the circle drawn at the tip of the second hand? What is the diameter?		
21	Why are wheels made round? (Hint: What would happen if the wheel was a square? What would happen to the distance from the center of the circle to the ground?)		
22	A round coaster, 4.5 cm long from center to edge. (a) Radius of coaster = ? (b) Diameter of coaster = ?		

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
23	$r = 3 \text{ cm} \cdot d = ?$		
24	$d = 22 \text{ cm} \cdot r = ?$		

25	Draw a circle with radius 2 cm, compass opening = ?		
26	The sides of the square are 4 cm long, and the largest circle is drawn inside. Diameter of circle = ?		
27	The hour hand of the clock is 5 cm long. The distance between the tip of the hour hand and the center of the circle is 5 cm. What is this length of the clock face? A. Diameter B. Radius C. Chord D. Circumference		

Advanced layer (total 3 questions, choose do)

#	Question	Difficulty	Working Space
28	The diagonal length of a square is 14 cm. What is the diameter of its circumcircle (the circle passing through its four vertices)? What is the radius?		
29	To draw a circle with a diameter of 16 cm, what should be the opening of the compass? If the maximum opening of the compass is only 6 cm, can this circle be drawn? Why?		
30	The diameter of a circle is 1 m and what is its radius in cm? (1 m = 100 cm)		

challenge layer (total 2 questions, choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
31	A rectangle is 14 cm long and 8 cm wide. Draw the largest circle within the rectangle. (a) What are the diameter and radius of this circle? (b) If the radius of this circle is increased by 1 cm, can it still be drawn within the rectangle? Why?		
32	Two concentric circles (sharing center O). The radius of the large circle is $R = 10$ cm, and the radius of the small circle is $r = 6$ cm. (a) How much longer is the diameter of the larger circle than the diameter of the smaller circle? (b) What is the width of the annular area between the large circle and the small circle in cm?		

IV. Class after homework

Basic must-do questions (total 5 questions)

#	Question	Difficulty	Working Space
H1	Radius $r = 3.5$ cm, diameter $d = ?$		
H2	Diameter $d = 9$ cm, radius $r = ?$		
H3	Draw a circle of diameter 6 cm with compass opening = ? (Must write down calculation steps)		
H4	The radius of a circle is 8 cm, its diameter is the longest _____ (fill in "chord", "radius" or "circumference"), and its length is _____ cm.		
H5	The sides of the square are 5 cm long and the largest circle is drawn inside. Diameter of circle = ?		

Advanced choose doquestion (total 3 questions,  choose do)

#	Question	Difficulty	Working Space
H6	The diagonal length of a square inscribed in a circle is 16 cm. What is the radius of this circle?		
H7	The hour hand of the clock is 6 cm long and the minute hand is 9 cm long. (a) Which needle has its tip farther from the center of the circle? How far away? (b) What is the difference in diameter of the circles drawn by the tips of the hour and minute hands?		
H8	A rope forms an exact circle and the distance from the center of the circle to the rope is 2 m. (a) What is the diameter of the circle enclosed by this rope? (b) If the rope is straightened and used as the diameter, how long does the rope need to be? (Tip: The diameter is the diameter of the rope after it is straightened)		

V. The Lesson core common errors summary

 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

☒ 本堂自我檢查 (完成後打剔)

☐ 我識得分辨每個知識點嘅陷阱

☐ 我能夠獨立完成  基礎題

☐ 我能夠挑戰  進階題

☐ 我記得住口訣

#	common error	Correct Approach
1	Confusing radius and diameter SEP : Misremembering $d = r \div 2$: Misremembering $d = r \div 2$	The diameter is longer than the radius: $d = r \times 2$; $r = d \div 2$
2	Set the compass opening to diameter SEP : To draw a circle with $d=8\text{cm}$, adjust the opening to 8cm : To draw a circle with $d=8\text{cm}$, adjust the opening to 8cm	Compass opening = radius ! First calculate $r = d \div 2 = 4\text{cm}$
3	Forget that all radii within the same circle are equal	Within the same circle, the length of any radius is equal (definition of a circle)
4	Think that the longest chord is not the diameter	Chord passing through center = diameter, is the longest chord in the circle
5	The units are not unified SEP : m and cm are mixed and not converted : m and cm are mixed without conversion	Unify the units before calculation: $1\text{m} = 100\text{cm}$
6	The diameter of the inscribed circle is the "length" of the rectangle	Diameter of the largest inscribed circle = length of the shorter side
7	The steps for drawing a circle are incomplete SEP : directly set the compass opening to the diameter value : Directly set the compass opening to the diameter value	Steps: ① Calculate r ② Adjust compass opening = r ③ Determine the center of the circle ④ Draw a circle

家長角落 · Parent Corner

今日學咗咩？小朋友學咗「**round understanding**」。

最易錯嘅 3 個陷阱：留意老師圈出嘅陷阱題目

你可以問小朋友：「你可唔可以解釋「round understanding」嘅最重要口訣俾我聽？」

溫馨提示：唔需要識教數學 — 只需要問小朋友「點解咁計？」同「有冇檢查陷阱？」就夠。

 教師參考：熱身題答案 → 1)___ 2)___ 3)___ 4)___ 5)___ | 課堂練習重點關注題號： 挑戰層 17-21

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 Related topics: L21 Understanding of circles · L22 Advanced area of circles · L06 Circumference and area of circles